

Chapter

Locus

1. Two straight lines PQ and PK cross each other at P at an angle of 75° . S is a stone on the road PQ . 800 m from P towards Q . By drawing a figure to scale $1 \text{ cm} = 100 \text{ m}$, locate the position of a flag staff X , which is equidistant from P and S , and is also equidistant from the road.

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2. By using ruler and compass only, construct a quadrilateral $ABCD$ in which $AB = 6.5 \text{ cm}$, $AD = 4 \text{ cm}$ and $\angle DAB = 75^\circ$. C is equidistant from the sides AB and AD , also C is equidistant from the points A and B .

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Solution

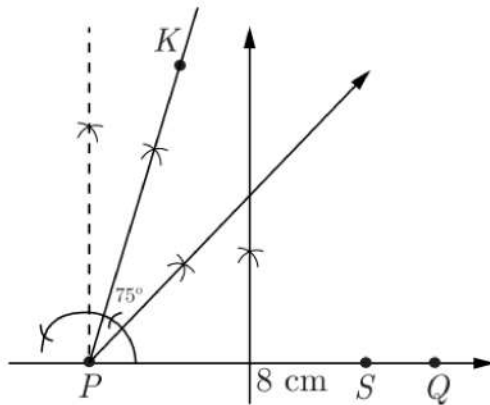
1.

Ans:

Steps of Construction:

1. Draw two straight lines PQ and PK intersecting at point P such that $\angle QPK = 75^\circ$.
2. On line PQ , mark a point S such that $PS = 8$ cm (since 800 m = 8 cm).
3. Draw the perpendicular bisector of the line segment PS .
4. Draw the angle bisector of $\angle KPS$.
5. Let the perpendicular bisector of PS and the angle bisector of $\angle KPS$ intersect at point X .

Point X is the required position of the flag staff, which is equidistant from P and S and also equidistant from the roads PQ and PK .



2.

Ans:

Steps of Construction:

1. Draw a line segment $AB = 6.5$ cm.
2. At point A , draw a ray making an angle of 75° with AB and cut off $AD = 4$ cm on this ray.

3. Draw the angle bisector of $\angle DAB$.
4. Draw the perpendicular bisector of AB to intersect the angle bisector at point C .
5. Join CB and CD .

Thus, $ABCD$ is the required quadrilateral.

